Exercises

Exercise 2.1 · Modulus and argument. Determine the modulus and an argument of: $1-i$; $-\sqrt{3}+i$; -5; $4i$.

Solution. $1-i$: $\sqrt{2}$, $-\frac{\pi}{4}$. $-\sqrt{3}+i$: 2, $\frac{5\pi}{6}$ (cosine $-\frac{\sqrt{3}}{2}$, sine $\frac{1}{2}$). -5: 5, π. $4i$: 4, $\frac{\pi}{2}$.

Exercise 2.2 · From the trigonometric form to the algebraic form. Write in algebraic form: $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$; $\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right)$; $6\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)$.

Solution. $2\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)=-1+i\sqrt{3}$; $\sqrt{2}\left(\frac{\sqrt{2}}{2}-i\frac{\sqrt{2}}{2}\right)=1-i$; $6\left(-\frac{\sqrt{3}}{2}+\frac{i}{2}\right)=-3\sqrt{3}+3i$.

Exercise 2.3 · Computing a modulus without expanding. Compute the modulus of $(1+i)^6\left(\sqrt{3}-i\right)$, then that of $\dfrac{3+4i}{1-2i}$.

Solution. $\left|(1+i)^6(\sqrt{3}-i)\right|=\left(\sqrt{2}\right)^6\times 2=8\times 2=16$. Next $\left|\dfrac{3+4i}{1-2i}\right|=\dfrac{5}{\sqrt{5}}=\sqrt{5}$.

Exercise 2.4 · Sets of points. Determine and draw the set of points M with affix z such that:

1. $|z-2+i|=3$;
2. $|z-1|=|z+i|$;
3. $|\overline{z}+2i|=1$.

Solution.

1. $|z-(2-i)|=3$: circle with centre the point of affix $2-i$ and radius 3.
2. With $A(1)$ and $B(-i)$, $AM=BM$: perpendicular bisector of $[AB]$. Setting $z=x+iy$: $(x-1)^2+y^2=x^2+(y+1)^2$, that is $y=-x$.
3. $\overline{z}+2i=\overline{z-2i}$, so $|\overline{z}+2i|=|z-2i|$: circle with centre the point of affix $2i$ and radius 1.

Exercise 2.5 · Nature of a triangle. Consider the points A, B, C with affixes $a=1+i$, $b=3+2i$ and $c=2+4i$. Compute AB, BC and AC, then determine the nature of the triangle ABC.

Solution. $AB=|2+i|=\sqrt{5}$, $BC=|-1+2i|=\sqrt{5}$, $AC=|1+3i|=\sqrt{10}$. Since $AB=BC$ and $AB^2+BC^2=10=AC^2$, the triangle is right-angled and isosceles at B (converse of Pythagoras' theorem).

Exercise 2.6 · Numbers of modulus 1.

1. Prove that if $|z| = 1$, then $z + \dfrac{1}{z}$ is real.
2. Let z and z' be two elements of $\mathbb{U}$ such that $zz' \neq -1$. Prove that $\dfrac{z + z'}{1 + zz'}$ is real.

Solution.

1. $\dfrac{1}{z} = \overline{z}$, so $z + \dfrac{1}{z} = z + \overline{z} = 2\,\mathrm{Re}(z)$, which is real.
2. The conjugate is $\dfrac{\overline{z} + \overline{z'}}{1 + \overline{z}\,\overline{z'}} = \dfrac{\frac{1}{z} + \frac{1}{z'}}{1 + \frac{1}{zz'}}$. Multiplying numerator and denominator by zz' gives $\dfrac{z' + z}{zz' + 1}$: the number equals its conjugate, so it is real.

Exercise 2.7 · A sequence of complex numbers. Let $z_0 = 4$ and, for every natural number n, $z_{n+1} = \dfrac{1 + i}{2}\, z_n$.

1. Compute z_1, z_2 and plot the corresponding points.
2. Prove that $(|z_n|)$ is a geometric sequence; give its common ratio and its general term.
3. Determine the limit of $|z_n|$. Interpret geometrically.
4. Determine the smallest integer n such that $|z_n| < 0.01$.

Solution.

1. $z_1 = 2 + 2i$ and $z_2 = \dfrac{(1+i)(2+2i)}{2} = (1 + i)^2 = 2i$.
2. $|z_{n+1}| = \left|\dfrac{1+i}{2}\right| |z_n| = \dfrac{\sqrt{2}}{2}\,|z_n|$: a geometric sequence with common ratio $\dfrac{\sqrt{2}}{2}$ and first term 4, so $|z_n| = 4\left(\dfrac{\sqrt{2}}{2}\right)^n$.
3. $0 < \dfrac{\sqrt{2}}{2} < 1$, so $|z_n| \to 0$: the points get closer to O while turning (by an eighth of a turn at each step, as chapter 3 will justify).
4. $4\left(\dfrac{\sqrt{2}}{2}\right)^n < 0.01$ is equivalent to $n \ln \dfrac{\sqrt{2}}{2} < \ln 0.0025$, that is $n > \dfrac{\ln 0.0025}{\ln(\sqrt{2}/2)} \simeq 17.29$. The smallest integer is 18: $|z_{17}| \simeq 0.0110$ and $|z_{18}| = \dfrac{4}{512} \simeq 0.0078$.

Problem

Problem · Triangle inequality and the case of equality.

1. Let z be a complex number. Prove that $\mathrm{Re}(z) \leq |z|$, with equality if and only if z is a nonnegative real number.
2. Let z and z' be two complex numbers. Prove that $|z+z'|^2 = |z|^2 + |z'|^2 + 2\,\mathrm{Re}\left(z\overline{z'}\right)$.
3. Deduce the triangle inequality $|z+z'| \leq |z| + |z'|$.
4. Assume $z' \neq 0$. Prove that equality $|z+z'| = |z| + |z'|$ holds if and only if there is a real number $\lambda \geq 0$ such that $z = \lambda z'$. Interpret with the images M and M' of z and z'.
5. Application. Consider the points M with affix z such that $|z-3-4i| \leq 2$.
   (a) Prove that for these points, $3 \leq |z| \leq 7$.
   (b) Determine the affixes of the points for which $|z| = 7$, then $|z| = 3$.

Solution · Problem.

1. With $z = a + ib$: $\mathrm{Re}(z) = a \leq |a| = \sqrt{a^2} \leq \sqrt{a^2+b^2} = |z|$. Equality forces $b = 0$ and $a = |a|$, so z is a nonnegative real number; the converse is immediate.
2. $|z+z'|^2 = (z+z')\left(\overline{z}+\overline{z'}\right) = z\overline{z} + z'\overline{z'} + z\overline{z'} + \overline{z}z'$. Now $\overline{z}z' = \overline{z\overline{z'}}$, so $z\overline{z'} + \overline{z}z' = 2\,\mathrm{Re}\left(z\overline{z'}\right)$.
3. By 1, $\mathrm{Re}\left(z\overline{z'}\right) \leq \left|z\overline{z'}\right| = |z|\,|z'|$. So $|z+z'|^2 \leq (|z| + |z'|)^2$, and both numbers compared are nonnegative.
4. Equality is equivalent to $\mathrm{Re}\left(z\overline{z'}\right) = \left|z\overline{z'}\right|$, that is, by 1, to $z\overline{z'} = \mu$, a nonnegative real number. In that case, $z = \frac{\mu}{|z'|^2}z'$ with $\lambda = \frac{\mu}{|z'|^2} \geq 0$. Conversely, if $z = \lambda z'$ with $\lambda \geq 0$, $z\overline{z'} = \lambda\,|z'|^2 \geq 0$. Geometrically: M lies on the ray $[OM')$.
5. (a) Let $w = 3+4i$, whose modulus is 5. On the one hand $|z| = |(z-w)+w| \leq 2+5 = 7$. On the other hand $5 = |w| = |(w-z)+z| \leq |z-w| + |z| \leq 2 + |z|$, so $|z| \geq 3$.
   (b) $|z| = 7$ requires equality in $|(z-w)+w| \leq |z-w| + |w|$ and $|z-w| = 2$: $z - w = \lambda w$ with $\lambda \geq 0$ and $|\lambda w| = 2$, that is $\lambda = \frac{2}{5}$ and $z = \frac{7}{5}w = \frac{21}{5} + \frac{28}{5}i$. Likewise, $|z| = 3$ gives $w - z = \frac{2}{5}w$, that is $z = \frac{3}{5}w = \frac{9}{5} + \frac{12}{5}i$. Both points do satisfy $|z-w| = 2$.

Teacher note · Status of the problem. The triangle inequality and its case of equality are a "possible problem" of the syllabus. Question 5 is easy to visualise: the disc with centre $(3,4)$ and radius 2 lies between the circles with centre O and radii 3 and 7, which it touches on the line joining O to the centre of the disc.